

Positive-Case Count and Synthetic Data Fidelity: A CART-Based Analysis of MIMIC-IV Clinical Data

William Chen
University of California, Irvine
Instructor: Prof. Sam Behseta
DATA298P

Abstract

A logistic fit’s precision is set by the number of positive cases rather than by the sample size. This report asks whether the same threshold governs a generator. CART-based synthesis (**synthpop**) is run on MIMIC-IV ($n = 70,954$ ICU stays) under two sweeps that hold the generator fixed and vary only the input: one raises the rare stratum’s positive count k at fixed sample size, the other raises the sample size n at fixed k . Together they separate count from prevalence.

The threshold does carry over, but the generator adds none of its own. Raising k drives divergence to zero between counts of forty and eighty, and over a matched eightfold increase it cuts interquartile width by 42 to 66% in all six series, while the same increase in n moves neither. The real arm’s divergence rate and a logistic fit to the same subsamples never involve the synthesizer, so the threshold belongs to the stratum and **synthpop** inherits it rather than imposing a second one. Synthesis adds a bias of its own that no count removes: in a companion simulation with a known truth, the synthetic median recovers 99% of an interaction of 0.5 but only 41% of one of 3. The finding transfers as a mechanism and not as a remedy.

A 500-row subsample carries 6 to 7% power against the interactions MIMIC-IV offers, and the smallest one it could detect is seven to twelve times the one present. Two recommendations follow: establish that a target effect is estimable in the source data before attributing any discrepancy to the generator, and calibrate variable-selection diagnostics against a permutation null, which here moves from 0.00 to 0.32 with class balance alone.

1 Introduction

That a logistic fit’s precision is set by the event count is established [1, 2], and separation at thin counts has been characterised since [3]. In earlier coursework I found the same under Bayesian logistic regression on the **birthwt** dataset: posterior stability depends on a predictor’s positive-case count rather than on overall sample size. Nobody has shown whether the same threshold governs a *generator*. Whether a CART-based synthesizer preserves a rare-stratum interaction is a question about which variables a tree spends a split on, not about the curvature of a likelihood, and the two need not share a threshold.

Synthetic data shares clinical records without exposing patients, and is usually judged by general utility, how closely the synthetic joint distribution resembles the real one [9]. A generator can score well on that and still lose the one coefficient a study was designed around, so what follows holds the generator fixed and measures a single interaction rather than overall similarity. It matters because of a specific misreading. A researcher handed a synthetic release of MIMIC-IV, finding the CHF by creatinine interaction absent from it, would reasonably conclude that the synthesizer had destroyed a clinical interaction. In the 500 rows in front of them that effect was never detectable to begin with, and the same discrepancy would have appeared with no synthesizer involved at all. Code is available at [GitHub Repository](#).

2 Data

2.1 Cohort

The cohort is MIMIC-IV v2.2 [8], restricted to hospital admissions with an associated ICU stay, one row per stay. Dropping the 3% with no first-day creatinine leaves a pool of 70,954, from which all subsampling draws. That 3% is not a random slice: a stay ending within hours has no time for the lab draw, so the pool leans toward stays that lasted long enough to be measured. The rows that remain repeat patients, so standard errors on the full-pool estimate are clustered.

2.2 Variables

Three predictors enter every fit: age, first-day maximum creatinine, and one comorbidity flag. Comorbidities are simplified ICD-9/ICD-10 prefix flags rather than a validated index such as Elixhauser or Charlson, since the design needs a known positive count. The three least prevalent of the five screened were carried forward: COPD (11%), CHF (28%) and Diabetes (31%). Even the rarest is too common to supply a thin stratum, so it is manufactured by subsampling. Creatinine is capped at the 99.5th percentile and left untransformed, so every creatinine coefficient below is per mg/dL.

3 Methods

3.1 Experimental Design

Both sweeps draw stratified subsamples and synthesize each with `synthpop::syn()` [7]. One model is fitted to the real subsample and again to the synthetic table, giving paired estimates $\hat{\beta}_{\text{real}}$ and $\hat{\beta}_{\text{syn}}$ of the interaction at $M = 100$ Monte Carlo replicates per grid point:

$$\text{hospital_expire_flag} \sim \text{age} + \text{creatinine} + \text{comorbidity} + \text{comorbidity:creatinine}$$

The distance between the two estimates is the fidelity measure, and the same subsamples carry over to the tree diagnosis, so one draw supports both analyses in Section 3.2. Both arms come from the same subsample, so the pool’s selection cancels in the comparison. Each sweep holds one input fixed and varies the other (Table 1). The count sweep moves prevalence along with the count; the size sweep moves prevalence alone, holding the count at 20, where the interaction draws its information mostly from the positive stratum and rests on the deaths inside twenty rows however large the subsample grows around them.

Sweep	Held fixed	Varied	What moves
Count sweep	$n = 500$	$k \in \{5, 10, 20, 40, 80, 150, 250\}$	count and prevalence together
Size sweep	$k = 20$	$n \in \{250, 500, 1000, 2000, 4000\}$	prevalence only, 8% to 0.5%

Table 1: The two sweeps. Each holds the generator fixed and varies one input.

Throughout, a *series* is one comorbidity in one arm, six in all, and a *cell* is one series at one grid point.

3.2 Analyses

Two analyses run on every grid point. The first compares $\hat{\beta}_{\text{real}}$ against $\hat{\beta}_{\text{syn}}$ on the median and interquartile width of the converged estimates, and on how often a fit fails at all (Section 3.3). The synthetic arm is fitted the way a data recipient would fit it: one synthetic copy, an ordinary GLM. The second refits the same subsamples with `rpart` under `synthpop::syn.cart`’s control parameters and records whether the comorbidity earns a split; a likelihood-ratio test dropping the comorbidity and its interaction asks a GLM the same question. Every configuration is re-run with the comorbidity shuffled, which holds the positive count fixed but destroys its link to the outcome, giving the null.

3.3 Two Failure Modes

Fits at the smallest counts fail often enough that the rates are reported as a result, not used as a filter. They fail in two ways. The first is **inestimable**: the table holds no positives, so the interaction returns NA. The second is **divergent**: the term is returned but runs past $|2|$, forty times the full-pool estimate, marking a likelihood maximized at the boundary [5].

Only the second appears in the real arm, which holds exactly k positives by construction, while the synthetic table generates its own column and can come back empty. Rates are shares of all M replicates, so converged, inestimable and divergent sum to one, and every quantile below uses converged fits only.

3.4 The Full-Pool Reference

MIMIC-IV carries no known ground truth, so the interaction fit on the whole pool stands in as the reference, $\hat{\beta}_{\text{full}}$, and every subsample estimate is read against it. All three are real, differ in sign, and run roughly three to four times smaller than the creatinine slope they modify (Table 2).

Comorbidity	Creatinine	$\hat{\beta}_{\text{full}}$	SE	p
CHF	0.208	-0.0497	0.0141	4.4×10^{-4}
Diabetes	0.234	-0.0709	0.0145	1.1×10^{-6}
COPD	0.188	+0.0668	0.0232	4.0×10^{-3}

Table 2: The interaction on the full pool, against the creatinine slope it modifies. Standard errors clustered on patient.

3.5 What This Design Can and Cannot Resolve

The subject here is what a generator does to a small subsample, so n is set by the question rather than by what the data could support. Measuring what that choice costs is what fixes the scale everything below is read on. At $n = 500$ the smallest interaction detectable at 80% power runs seven to twelve times the one the pool holds (Table 3), and reaching these effects would take most of the pool.

Comorbidity	Smallest $ \beta $ detectable	Factor	Rows needed
CHF	0.47	9.5×	45,000
Diabetes	0.49	6.8×	23,000
COPD	0.78	11.6×	67,000

Table 3: The detection floor at 80% power. An interaction must reach the first column before a 500-row subsample finds it four times in five; the factor compares that threshold with $\hat{\beta}_{\text{full}}$ in Table 2.

No subsample this size can say whether synthesis recovers a true interaction, so the sweeps do not ask. They measure precision and failure behaviour, readable at any effect size, and leave bias to a companion simulation where the truth is known (Section 4.3).

4 Results

4.1 Count, Not Sample Size, Removes Divergence

Count sweep: divergence stops by a count of eighty. Divergence runs at 18 to 40% at $k = 5$ and every series is at zero by $k = 80$. Not at one shared count: two are there already at $k = 40$, and a zero over 100 replicates bounds a rate at 3.6% rather than proving it nil, so the threshold sits in $(40, 80]$ for each series. Neither prevalence nor the sign of the interaction moves it.

Size sweep: divergence never stops. No sample size does the same. Over a matched eightfold increase in n divergence stays between 2 and 24% and no series reaches zero. The arm separates cleanly here and the size does not: the synthetic rate is higher in fourteen of the fifteen comorbidity-by-size comparisons, and no sample size closes the gap.

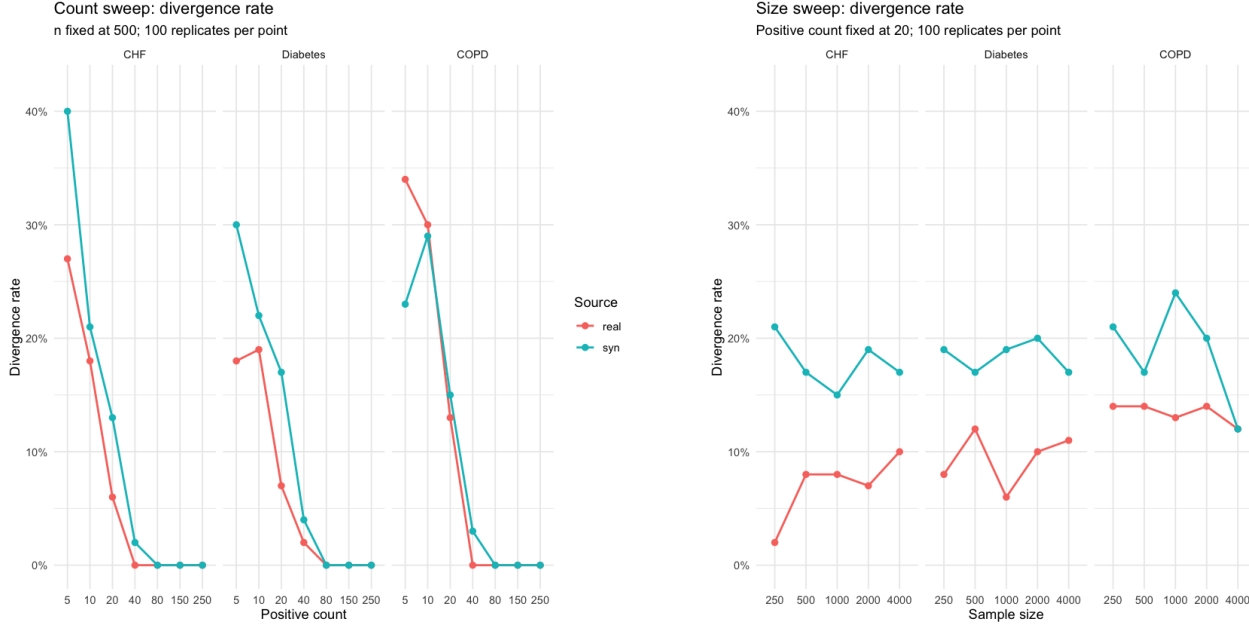


Figure 1: Left, the count sweep: divergence falls away. Right, the size sweep: it does not. Common scale; the two smallest counts understate the true rate.

Checking the $|2|$ rule. The $|2|$ threshold is one this study set, not one the data supplies, so it is checked against a separation test. It understates divergence at the smallest counts: a penalized fit [4] finds 64 to 76% of real fits separated at $k = 5$ where $|2|$ flags 18 to 34%. The two agree from $k = 20$ up, and the penalized fit does not move the count at which divergence stops. The rule mismeasures how bad the smallest counts are, not where failure ends.

4.2 The Count Narrows the Spread; the Sample Size Does Not

Count sweep: the count narrows it, and it is still four times too wide. Every series loses 55 to 69% of its width from $k = 20$ to $k = 250$. The narrowest cell still runs 0.194 wide against a reference of 0.0497 (Table 2), nearly four times the quantity being estimated. What low counts cost is precision and not location: the middle half of the estimates still contains $\hat{\beta}_{\text{full}}$ in 39 of the 42 cells.

Size sweep: no sample size narrows it. Over a matched eightfold window, n from 500 to 4000 against k from 20 to 150, the sample size moves width between -19% and $+10\%$ with no consistent sign, while the count over that shorter window still cuts it by 42 to 66% in all six series.

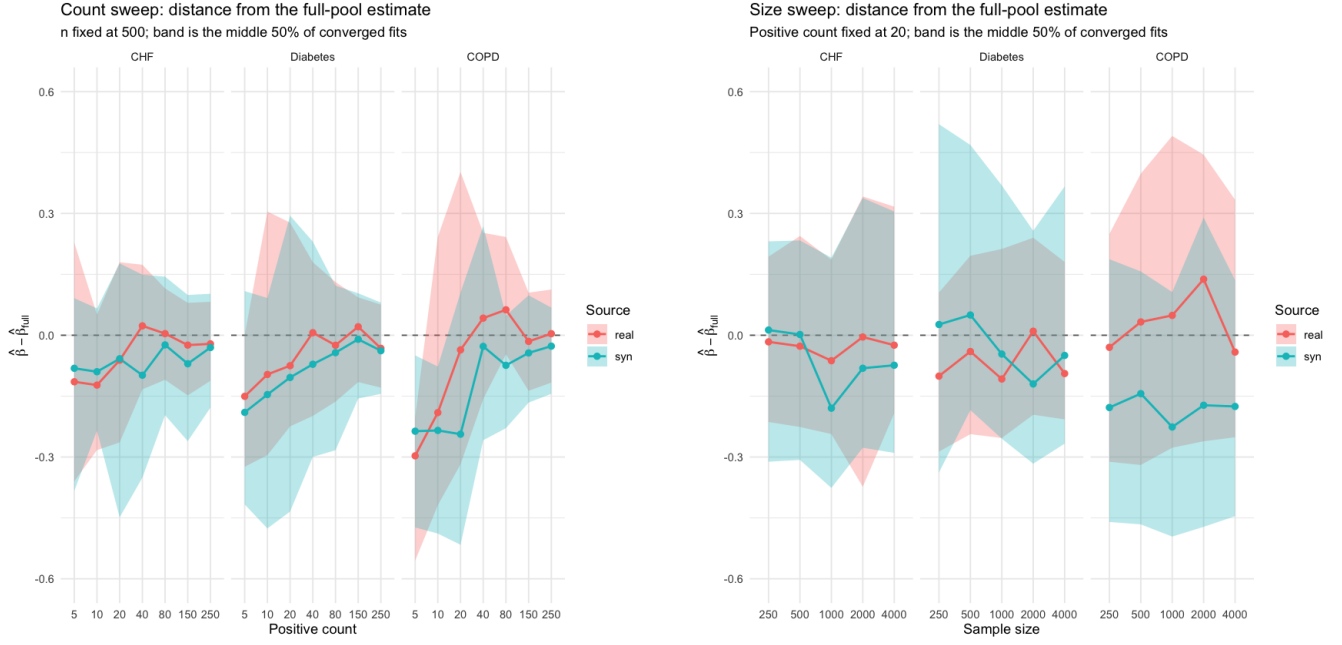


Figure 2: Left, the count sweep: the spread narrows. Right, the size sweep: it does not. Common scale, plotted as distance from $\hat{\beta}_{full}$; band is the middle 50% of converged fits.

The rate explains the real arm, not the synthetic one. An interaction's variance scales as $1/k + 1/(n - k)$. Rescaled by that rate the real width tracks at 0.97, and under the size sweep the rate predicts almost no change, which is what the data shows. The synthetic arm sits above it at 1.14, wider than the real arm in all nine comorbidity-by-count comparisons (Figure 3; sign test $p = 0.004$).

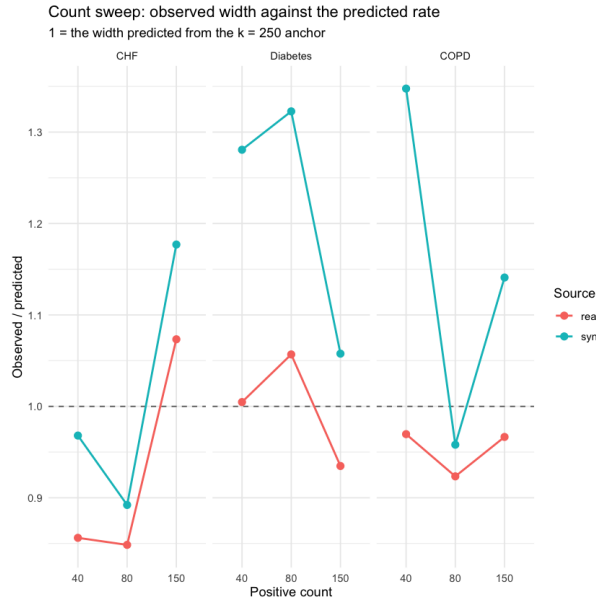


Figure 3: The real arm tracks the width its count predicts; the synthetic arm sits above the real one in all nine comparisons.

A sign the sample size does not repair. COPD is the one comorbidity whose interaction is positive, $+0.0668$, and the count sweep recovers that sign once k passes 20. Holding the count at 20 does not: the synthetic median stays negative at all five sample sizes, -0.077 to -0.159 . Sixteen times the data leaves the sign where twenty positives put it.

4.3 Synthesis Attenuates the Interaction, and No Count Repairs It

Where MIMIC-IV has no truth to compare against, a simulated population does. It is generated from a logistic model with a known `chf` $\times$ lactate interaction, then put through the same pipeline: a subsample drawn at each count, synthesized, and both arms fitted as before. The three true values sit well above the floor of Section 3.5.

True value	Median estimate		Median error	
	Real	Synthetic	Real	Synthetic
0.5	0.521	0.496	0.164	0.219
1.5	1.73	0.891	0.292	0.713
3.0	2.75	1.22	0.688	1.90

Table 4: Estimates at $k = 250$ against the true value. The last two columns are the median of $|\hat{\beta} - \text{true}|$ over replicates.

The larger the interaction, the more is lost. Synthesis keeps 99% of a true value of 0.5, 59% of 1.5 and 41% of 3 (Table 4), and no count on the grid closes the gap (Figure 4). The row at 0.5 is not the exception it looks like: an estimator that pulls toward zero is accurate against any target near zero, and the error columns show what the medians hide, 34% larger on the synthetic side where the medians agree to within 1%.

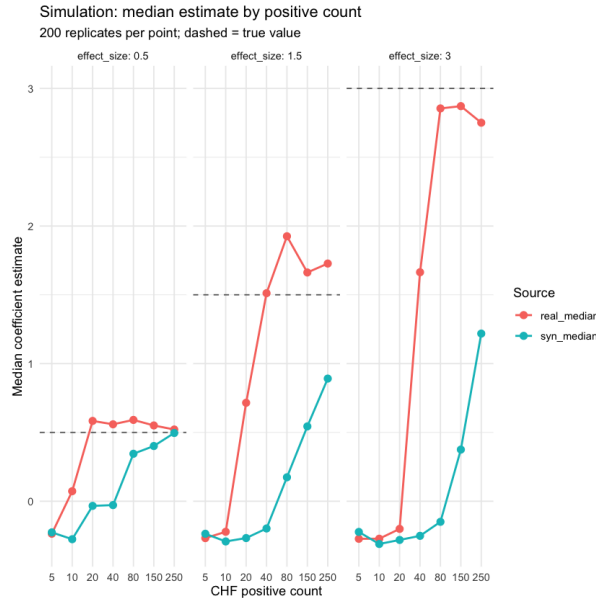


Figure 4: The real arm converges on the truth; the synthetic arm lags at every count.

The synthetic arm diverges less, not more. At a true value of 3 the real arm diverges in 21 to 29% of replicates while the synthetic arm diverges in 2.5 to 10.5%, the reverse of the MIMIC-IV pattern. A strong interaction puts the real subsample close to separation; the tree smooths that away, so the synthetic fit converges more readily and converges on the wrong value. The attenuation is bought, not incurred.

4.4 The Split Diagnostic Is Unreadable Without Its Null

What looks like recovery in the count sweep is class balance, not signal. The diagnostic rises from 0 at $k = 5$ to 0.26 to 0.40 at $k = 250$, but the permutation null rises with it, to as high as 0.32 (Figure 5, left): the column grows from 1% positive to 50%, and a balanced binary variable is easy to split on whether or not

it carries signal [10]. Pooled across the grid the signal is still there, an odds ratio of 2.0 ($p = 6 \times 10^{-6}$) from a conditional logistic model stratified on (comorbidity, replicate) pairs, narrowing by 0.81 per doubling of the count ($p = 0.015$) as the null closes in.

Holding the count at 20 runs balance the other way, down to 0.5% positive at $n = 4000$. The null falls with it to near zero while the observed rate stays put (Figure 5, right), and with the null out of the way the signal is legible without pooling: 104 discordant replicates to 32. The diagnostic is readable where its null is near zero, which here means low prevalence.

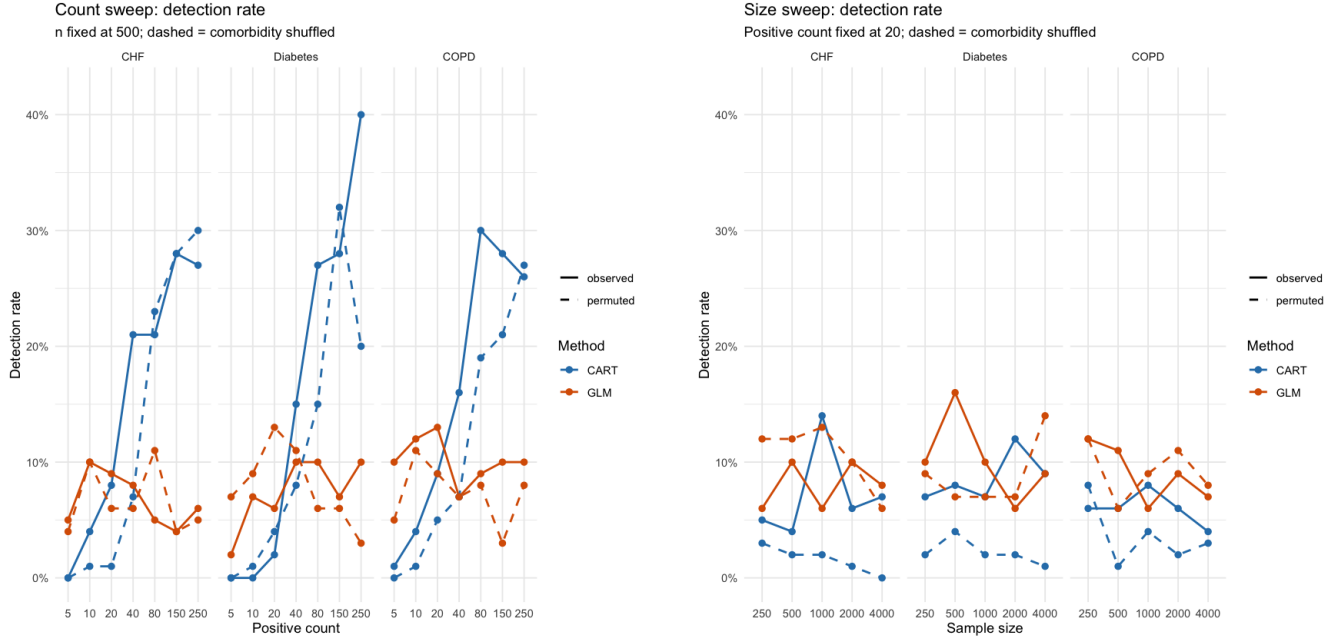


Figure 5: Both arms here are real; dashed is the same column shuffled. Left, the count sweep: the null climbs alongside the observed rate, so the rise is not recovery. Right, the size sweep: the null falls away and the observed rate stays put.

5 Discussion

The design, not the generator. The sweeps returned the floor of Section 3.5. Theory gives a 500-row subsample 6 to 7% power against these interactions and the GLM arm returned 6.7 to 10.1%, both within a few points of the 5% a powerless test would produce, which locates the failure in the design and not in the pipeline. Neither sweep implicates CART on its own: the GLM does no better on real labels than on shuffled ones, and CART is only the method that fails visibly, by declining to spend a split.

A threshold, but not a cutoff. The count at which divergence stops sits in $(40, 80]$ for each series, but at different points inside. That is the expected shape, not a ragged result, since van Smeden et al. find no rationale for a fixed events-per-variable criterion [2]. A generator inherits a dependence on the count, not a universal cutoff.

Three readings that disagree. Failure, spread and location do not move together. The synthetic arm spreads wider than its own count predicts and sits below the truth, yet at a large true effect it fails less often than the real one. Read on its own that last reading looks like stability, and stability is what the smoothing buys.

A privacy-fidelity tension. CART’s reluctance to isolate a subgroup with few supporting cases is not simply a limitation to work around. A tree willing to carve out a node of five patients would also be likelier

to overfit to and risk re-identifying them, and that same reluctance leaves the stratum missing at $k = 5$. The tension belongs to the method, not the implementation, though only its fidelity side is measured here.

Limitations. The diagnostic tree is a proxy for the generator: three predictors rather than the full feature set `synthpop` synthesizes jointly, matched on control parameters but not on visit order. Both are permissive, which makes non-selection the more reliable outcome. Monte Carlo error at $M = 100$ reaches 13% on interquartile widths, so a cell that misses significance is not evidence that nothing is there. The single-copy GLM carries no correction for the variance synthesis contributes [6], so the two arms are compared as procedures in use, not as matched estimators. One continuous variable and one outcome were examined, two of the five screened comorbidities were left unanalysed, and all results come from a single MIMIC-IV snapshot.

6 Conclusion

The positive count governs every quantity the two sweeps measure and sample size governs none: divergence falls to zero between counts of forty and eighty, interquartile width falls by 42 to 66% over a matched increase in the count, and neither CART’s use of the comorbidity nor a logistic fit to the same subsamples moves with n . These are four readings on the same replicates rather than four independent confirmations, but two of them never touch the synthesizer. In that narrow sense the finding from Bayesian logistic regression extends to CART-based synthesis.

The remedy does not carry over. The generator adds an attenuation no count removes, partial even where the target is large enough to reach, and at clinical effect sizes neither arm recovers the MIMIC-IV interactions at all: the narrowest cell still spans nearly four times the quantity being estimated.

For synthetic data evaluation, two things follow: establish that the target effect is estimable in the source data before attributing a discrepancy to the generator, and calibrate any variable-selection diagnostic against a permutation null. Neither is specific to `synthpop`: any generator that declines to model a subgroup it has too few cases to support should show the same threshold, and the two sweeps needed to find it require nothing beyond the source data and the generator itself.

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